

Computational Finance

Lecture 10 – Pricing Barrier Options

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Lecture 10 – Pricing Barrier Options

- The pricing problem
- Analytical results under GBM
- The COS method for pricing barrier options

Recall the Definition of Barrier Options

- An *up-and-out call* option has a payoff of zero **if** the asset crosses some pre-defined barrier $B > S_0$ at some time in $[0, T]$. If the barrier is not crossed, the payoff becomes that of a European call, $\max(S_T - K, 0)$.
- An *up-and-in call* option has a payoff of zero **unless** the asset crosses some pre-defined barrier $B > S_0$ at some time in $[0, T]$. If the barrier is crossed, the payoff becomes that of a European call, $\max(S_T - K, 0)$.
- There are also put versions of the above calls; just replace the word “call” by “put”. This gives a total of eight different up/down-and-in/out calls/puts.
- In each case, an analytical formula for the option value can be obtained.

- Parity relations between the prices of barrier options and vanilla calls/puts under the risk-neutral measure:

$$C_{\text{up-in}}(t) + C_{\text{up-out}}(t) = e^{-r(T-t)} \mathbb{E}[(S_T - K)^+ | \mathcal{F}_t],$$

$$C_{\text{dn-in}}(t) + C_{\text{dn-out}}(t) = e^{-r(T-t)} \mathbb{E}[(S_T - K)^+ | \mathcal{F}_t],$$

$$P_{\text{up-in}}(t) + P_{\text{up-out}}(t) = e^{-r(T-t)} \mathbb{E}[(K - S_T)^+ | \mathcal{F}_t],$$

$$P_{\text{dn-in}}(t) + P_{\text{dn-out}}(t) = e^{-r(T-t)} \mathbb{E}[(K - S_T)^+ | \mathcal{F}_t].$$

- Solve the expectation via the martingale approach:

$$V(x, t) = e^{-r(T-t)} \mathbb{E}_{X_t=x} \left[\left(\alpha \cdot (e^{X_T} - K) \right)^+ \cdot \mathbf{1}_{\{\tau_B > T\}} \right],$$

where $\mathbb{E}_{X_t=x}$ denotes an expectation under a diffusion process and

- B : the up-and-out barrier level;
 - τ_B : the barrier-breaching time;
 - α : +1 for a call and -1 for a put.
- Or the equivalent PDE – localized Feynman–Kac theorem: for $x \in [a, b]$,

$$\frac{\partial V(x, t)}{\partial t} + r(x, t) \frac{\partial V(x, t)}{\partial x} + \frac{1}{2} \sigma^2(x, t) \frac{\partial^2 V(x, t)}{\partial x^2} - r(x, t) V(x, t) = 0,$$

with terminal condition $V(x, T) = (\alpha \cdot (e^{X_T} - K))^+$ and boundary conditions $V(b, t) = 0$, $V(a, t) = 0$.

- Under GBM (Black–Scholes) there is a rich literature regarding:
 - the distribution of the running maximum;
 - the joint distribution of the Brownian motion and its running maximum;
 - the analytical solutions of various barrier options;
 - series-expansion-based solutions via the PDE.

The Expectation Route

- Define the running maximum

$$\Xi_t^T = \max_{\tau \in [t, T]} X_\tau.$$

- Then

$$\begin{aligned} V(x, t) &= e^{-r(T-t)} \mathbb{E}_{X_t=x} \left[(\alpha \cdot (e^{X_T} - K))^+ \cdot \mathbf{1}_{\{\tau_B > T\}} \right] \\ &= e^{-r(T-t)} \mathbb{E}_{X_t=x} \left[(\alpha \cdot (e^{X_T} - K))^+ \cdot \mathbf{1}_{\{\Xi_t^T \in [a, b]\}} \right] \\ &= e^{-r(T-t)} \int_{-\infty}^{\infty} \int_a^b (\alpha \cdot (e^y - K))^+ \cdot p(\xi, y) d\xi dy, \end{aligned}$$

where $p(\xi, y)$ is the joint density of Ξ_t^T and X_T .

- Hence the key is to derive (or estimate) the joint distribution of the underlying and its running maximum.

The Running Maximum of Brownian Motion

- Let W_t denote the standard Brownian motion started at $W_0 = 0$.
- Let Ξ_0^T denote the running maximum,

$$\Xi_0^T = \max_{t \in [0, T]} W_t.$$

- While it is well known that $W_T \sim \mathcal{N}(0, T)$, computing the distribution of Ξ_0^T might seem difficult.
- However, this is not the case, due to the [reflection principle](#).

The Reflection Principle (1/3)

- Since $W_0 = 0$, we have almost surely

$$\Xi_0^T = \max_{t \in [0, T]} W_t \geq 0.$$

- Given $a > W_0 = 0$, the first hitting time of level a by W_t is

$$\tau_a = \inf\{t \geq 0 : W_t = a\}.$$

- By spatial symmetry of Brownian motion,

$$\mathbb{P}(W_T \geq a \mid \tau_a \leq T) = \mathbb{P}(W_T > a \mid \tau_a \leq T) = \mathbb{P}(W_T \leq a \mid \tau_a \leq T) = \frac{1}{2}.$$

- Moreover, the events

$$\{\Xi_0^T \geq a\} = \{\tau_a \leq T\}$$

are equivalent.

The Reflection Principle (2/3)

- For $a > 0$,

$$\begin{aligned}\mathbb{P}(\tau_a \leq T) &= \mathbb{P}(\tau_a \leq T, W_T > a) + \mathbb{P}(\tau_a \leq T, W_T \leq a) \\ &= 2 \mathbb{P}(\tau_a \leq T, W_T \geq a) = 2 \mathbb{P}(\Xi_0^T \geq a, W_T \geq a) \\ &= 2 \mathbb{P}(W_T \geq a) = \mathbb{P}(W_T \geq a) + \mathbb{P}(W_T \leq -a) = \mathbb{P}(|W_T| \geq a),\end{aligned}$$

where we used $\{W_T \geq a\} \subset \{\Xi_0^T \geq a, W_T \geq a\} \subset \{W_T \geq a\}$.

- Thus the maximum has the same distribution as the absolute value:

$$\mathbb{P}(\Xi_0^T \leq a) = \mathbb{P}(\tau_a \geq T) = \mathbb{P}(|W_T| \leq a) = \frac{1}{\sqrt{2\pi T}} \int_{-a}^a e^{-\frac{x^2}{2T}} dx = \frac{2}{\sqrt{2\pi T}} \int_0^a e^{-\frac{x^2}{2T}} dx.$$

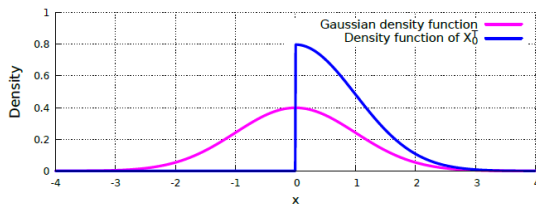
- Probability density:

$$p_{\Xi_0^T}(a) = \frac{d \mathbb{P}(\Xi_0^T \leq a)}{da} = \sqrt{\frac{2}{\pi T}} e^{-\frac{a^2}{2T}} \cdot \mathbf{1}_{[0, \infty)}(a), \quad a \in \mathbb{R}.$$

The Reflection Principle (3/3)

- As a consequence of the existence of the density $p_{\Xi_0^T}(a)$,

$$\mathbb{P}(W_t \leq 0, \forall t \in [0, \varepsilon]) = \mathbb{P}(\Xi_0^\varepsilon = 0) = \int_0^0 p_{\Xi_0^\varepsilon}(a) da = 0.$$



Probability density of the maximum Ξ_0^1 of Brownian motion over $[0, 1]$. Source: [Privault, MA5182 notes](#).

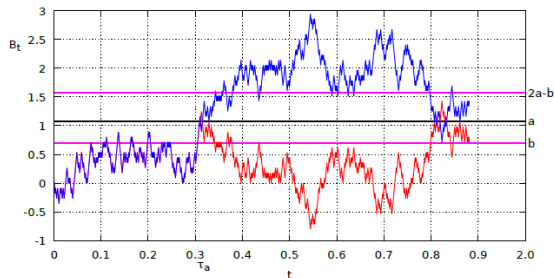
The Joint Probability Density of W_T and Ξ_0^T (1/3)

- For any $b \leq a$,

$$\mathbb{P}(W_T < b \mid \tau_a < T) = \mathbb{P}(W_T > a + (a - b) \mid \tau_a < T),$$

or equivalently,

$$\mathbb{P}(\Xi_0^T \geq a, W_T < b) = \mathbb{P}(\Xi_0^T \geq a, W_T > 2a - b).$$



Reflected Brownian motion with $a = 1.07$. Source: [Privault, MA5182 notes](#).

The Joint Probability Density of W_T and Ξ_0^T (2/3)

- Since $2a - b \geq a$,

$$\mathbb{P}(\Xi_0^T \geq a, W_T < b) = \mathbb{P}(\Xi_0^T \geq a, W_T > 2a - b) = \mathbb{P}(W_T > 2a - b),$$

where we used

$$\{W_T \geq 2a - b\} \subset \{\Xi_0^T \geq 2a - b, W_T > 2a - b\} \subset \{\Xi_0^T \geq a, W_T > 2a - b\} \subset \{W_T > 2a - b\}.$$

- This shows that $\{W_T \geq 2a - b\} = \{\Xi_0^T \geq a, W_T > 2a - b\}$, hence

$$\mathbb{P}(\Xi_0^T > a, W_T \leq b) = \mathbb{P}(W_T \geq 2a - b) = \frac{1}{\sqrt{2\pi T}} \int_{2a-b}^{\infty} e^{-\frac{x^2}{2T}} dx.$$

The Joint Probability Density of W_T and Ξ_0^T (3/3)

- The joint density is

$$\begin{aligned} p_{\Xi_0^T, W_T}(a, b) &= \frac{\partial^2}{\partial a \partial b} \mathbb{P}(\Xi_0^T \leq a, W_T \leq b) \\ &= \frac{\partial^2}{\partial a \partial b} (\mathbb{P}(W_0^T \leq b) - \mathbb{P}(\Xi_0^T > a, W_T \leq b)) \\ &= -\frac{\partial^2}{\partial a \partial b} \mathbb{P}(\Xi_0^T > a, W_T \leq b). \end{aligned}$$

- Plugging in the previous result,

$$p_{\Xi_0^T, W_T}(a, b) = \sqrt{\frac{2}{\pi}} \frac{2a - b}{T^{3/2}} e^{-\frac{(2a-b)^2}{2T}} \cdot \mathbf{1}_{\{a \geq \max(b, 0)\}}.$$

- For drifted Brownian motion, apply a change of measure.

Closed-Form Solution for Barrier Options Under GBM

- We can now price any exotic option defined on the maximum of Brownian motion, including barrier options:

$$\begin{aligned} V(x, t) &= e^{-r(T-t)} \mathbb{E} \left[\phi \left(S_T, \max_{\tau \in [t, T]} S_\tau \right) \right] \\ &= e^{-r(T-t)} \mathbb{E} \left[\phi \left(S_t e^{\sigma W_T + (r - \frac{1}{2} \sigma^2)(T-t)}, S_0 e^{\sigma \Xi_0^T + (r - \frac{1}{2} \sigma^2)(T-t)} \right) \right] \\ &= e^{-rT} \int_{-\infty}^{\infty} \int_{-\infty}^{\infty} \phi \left(S_t e^{\sigma y + (r - \frac{1}{2} \sigma^2)(T-t)}, S_0 e^{\sigma \xi + (r - \frac{1}{2} \sigma^2)(T-t)} \right) \cdot p(\xi, y) d\xi dy. \end{aligned}$$

- For an up-and-out call, the closed-form solution is

$$\begin{aligned} C_{\text{up-and-out}} &= S_t \left[N(d_1) - N(e_1) - (B/S_t)^{1+2r/\sigma^2} (N(f_2) - N(g_2)) \right] \\ &\quad - e^{-rT} K \left[N(d_2) - N(e_2) - (B/S_t)^{-1+2r/\sigma^2} (N(f_1) - N(g_1)) \right], \end{aligned}$$

with

$$\begin{aligned} e_{1,2} &= \frac{1}{\sigma \sqrt{T-t}} \left[\ln \frac{S_t}{B} + (r \pm \frac{1}{2} \sigma^2)(T-t) \right], \\ f_{2,1} &= \frac{1}{\sigma \sqrt{T-t}} \left[\ln \frac{S_t}{B} - (r \pm \frac{1}{2} \sigma^2)(T-t) \right], \\ g_{2,1} &= \frac{1}{\sigma \sqrt{T-t}} \left[\ln \frac{S_t K}{B^2} - (r \pm \frac{1}{2} \sigma^2)(T-t) \right]. \end{aligned}$$

The PDE Route (1/4)

- Use the [reflection principle](#) for the BS equation:
 - Suppose we have a solution to

$$\frac{\partial V}{\partial t} + rS \frac{\partial V}{\partial S} + \frac{1}{2}\sigma^2 S^2 \frac{\partial^2 V}{\partial S^2} - rV = 0.$$

- Then for any constant B (representing a barrier),

$$U(S, t) = S^{2\alpha} V\left(\frac{B^2}{S}, t\right), \quad \alpha = \frac{1}{2}\left(1 - \frac{2r}{\sigma^2}\right),$$

satisfies the same equation.

- Equivalent to applying the [method of images](#) to restore the usual boundary condition of the heat equation, after a change of variables on the BS PDE.
- This yields the analytical solution for a down-and-out call:

$$C_{\text{down-and-out}}(S_t, K) = C_{\text{vanilla}}(S_t, K) - \left(\frac{S}{B}\right)^{2\alpha} C_{\text{vanilla}}\left(\frac{B^2}{S_t}, K\right).$$

The PDE Route (2/4)

- Or use **Fourier series** to solve the reduced heat equation.
- Let $f : [a, b] \rightarrow \mathbb{R}$ be continuous with compact support, and let

$$\tau_{\notin[a,b]} := \inf\{s \geq t : X_s < a \text{ or } X_s > b\}$$

be the first time X_s exits $[a, b]$.

- Then

$$V(x, t) = e^{-r(T-t)} \mathbb{E} \left[f(X_T) \cdot \mathbf{1}_{\{\tau_{\notin[a,b]} > T\}} \mid X_t = x \right]$$

is the unique bounded solution of the BS pricing PDE

$$\frac{\partial V}{\partial t} + (r - \frac{1}{2}\sigma^2) \frac{\partial V}{\partial x} + \frac{1}{2}\sigma^2 \frac{\partial^2 V}{\partial x^2} - rV = 0, \quad x \in [a, b],$$

with the boundary and terminal conditions $V(a, t) = 0 = V(b, t)$, $V(x, T) = f(x)$, $x \in [a, b]$, where $f(x) = K(e^x - 1)^+$ for a call.

The PDE Route (3/4)

- Apply the change of variables:
 - $t \rightarrow \tau = \frac{1}{2}\sigma^2(T - t)$ and set $\kappa = 2r/\sigma^2$;
 - $V(x, \tau) = e^{\alpha x + \beta \tau} U(x, \tau)$ with $\alpha = -\frac{1}{2}(\kappa - 1)$ and $\beta = -\alpha^2 - \kappa$;
 - $z := x - a$.
- This yields the heat equation for $z \in [0, b - a]$ and $\tau \in [0, \frac{1}{2}\sigma^2 T]$:

$$\begin{cases} \frac{\partial U}{\partial \tau} = \frac{\partial^2 U}{\partial z^2}, \\ U(0, \tau) = 0, \\ U(b - a, \tau) = 0, \\ U(z, 0) = g(z) = e^{-\alpha(z+a)} K(e^{z+a} - 1)^+. \end{cases} \quad z \in [0, b - a],$$

The PDE Route (4/4)

- To solve the heat equation, compute the sine series for $g(z)$:

$$g(z) = \sum_{n=1}^{\infty} b_n \sin\left(\frac{n\pi z}{b-a}\right), \quad b_n = \frac{2}{b-a} \int_0^{b-a} g(z) \sin\left(\frac{n\pi z}{b-a}\right) dz.$$

- The solution to the heat equation is

$$U(z, \tau) = \sum_{n=1}^{\infty} b_n \sin\left(\frac{n\pi z}{b-a}\right) e^{-\left(\frac{n\pi}{b-a}\right)^2 \tau}.$$

- Hence the solution to the barrier-option pricing PDE is

$$V(x, t) = e^{-r(T-t)} \sum_{n=1}^{\infty} b_n \sin\left(\frac{n\pi(x-a)}{b-a}\right) e^{\alpha x} e^{-\frac{1}{2}\sigma^2(T-t)\left[(n\pi/(b-a))^2 + \alpha^2\right]}.$$

The COS Method

- Under Lévy processes
- Under Heston's model

Focus on Discretely-Monitored Barriers

- The payoff for a discretely-monitored up-and-out barrier option is

$$V(x, \min(T, \tau)) = (\max(\alpha(S_T - K), 0)) \cdot \mathbf{1}_{\tau > T} + R \cdot \mathbf{1}_{\tau \leq T},$$

with

$$\tau := \inf\{t_m \geq t_0 : S(t_m) \geq H\},$$

where $\alpha = +1$ for a call and $\alpha = -1$ for a put, R is a rebate, and $\mathbf{1}_A$ is the indicator function. In other words,

$$\mathbf{1}_{\tau > T} = \prod_{m=1}^M \mathbf{1}_{\{S(t_m) < H\}}.$$

- Discretely-monitored “in” barrier options are defined similarly.

Re-Write the Pricing Formula (1/3)

- Pricing formula:

$$V(X_0, t_0) = \mathbb{E}^{\mathbb{Q}}[V(X_M, t_M) \cdot \mathbf{1}_{\tau > T} + e^{-r\tau} R \cdot \mathbf{1}_{\tau \leq T} \mid X_0],$$

- $X_m = \ln(S(t_m)/K)$, $h = \ln(H/K)$, $m = M, M-1, \dots, 1$;
 - $V(X_M, t_M) = e^{-rt_M} \max(\alpha K(e^{X_M} - 1), 0)$;
 - $\tau := \inf\{t_m \geq t_0 : X_m \geq h\}$, so $\mathbf{1}_{\tau > T} = \prod_{m=1}^M \mathbf{1}_{\{X_m < h\}}$.
- Focus on the first part of the pricing formula and apply the tower property:

$$\begin{aligned}\mathbb{E}^{\mathbb{Q}}[V(X_M, t_M) \cdot \mathbf{1}_{\tau > T} \mid X_0] &= \mathbb{E}^{\mathbb{Q}}\left[V(X_M, t_M) \cdot \prod_{m=1}^M \mathbf{1}_{\{X_m < h\}} \mid X_0\right] \\&= \int_{\mathbb{R}^M} V(x_M, t_M) \prod_{m=1}^M \mathbf{1}_{\{x_m < h\}} f(x_M, x_{M-1}, \dots, x_1 \mid x_0) dx^M \\&= \int_{\mathbb{R}^M} V(x_M, t_M) \prod_{m=1}^M \mathbf{1}_{\{x_m < h\}} \\&\quad \cdot f(x_M \mid x_{M-1}, \dots, x_0) \cdots f(x_1 \mid x_0) dx^M.\end{aligned}$$

Re-Write the Pricing Formula (2/3)

- For a Markov process,

$$\begin{aligned}\mathbb{E}^{\mathbb{Q}}[V(X_M, t_M) \cdot \mathbf{1}_{\tau > T} \mid X_0] &= \int_{\mathbb{R}^M} V(x_M, t_M) \prod_{m=1}^M \mathbf{1}_{\{x_m < h\}} f(x_M \mid x_{M-1}) \cdots f(x_1 \mid x_0) dx^M \\ &= \int_{-\infty}^{+\infty} f(x_1 \mid x_0) \mathbf{1}_{\{x_1 < h\}} \cdots \\ &\quad \int_{-\infty}^{+\infty} f(x_{M-1} \mid x_{M-2}) \mathbf{1}_{\{x_{M-1} < h\}} \\ &\quad \int_{-\infty}^{+\infty} V(x_M, t_M) f(x_M \mid x_{M-1}) \mathbf{1}_{\{x_M < h\}} dx_M dx_{M-1} \cdots dx_1.\end{aligned}$$

Re-Write the Pricing Formula (3/3)

- Equivalently, write the pricing formula as a backward recursion (with rebate). For $m = M, M - 1, \dots, 2$:

$$c(x, t_{m-1}) = e^{-r(t_m - t_{m-1})} \int_{-\infty}^{+\infty} v(y, t_m) f(y | x) dy,$$

$$v(x, t_{m-1}) = \begin{cases} e^{-r(T - t_{m-1})} R, & x \geq h, \\ c(x, t_{m-1}), & x < h. \end{cases}$$

- $c(x, t_{m-1})$ is the *continuation value* at time t_{m-1} ;
 - $v(x, t)$ is the option value at time t ;
 - $x = \ln(S(t_{m-1})/K)$ and $y = \ln(S(t_m)/K)$ are the state variables;
 - $f(y | x)$ is the conditional density of y given x .
- For $m = 1$,

$$v(x, t_0) = e^{-r\Delta t} \int v(y, t_1) f(y | x) dy,$$

i.e. the price of the barrier option at the initial time.

Pricing Barrier Options by the COS Method (1/8)

- The main idea is to apply the COS method to value $c(x, t_{m-1})$ for $m = M, M - 1, \dots, 2$:

$$c(x, t_{m-1}) = e^{-r(t_m - t_{m-1})} \int_{-\infty}^{+\infty} v(y, t_m) f(y | x) dy,$$

$$v(x, t_{m-1}) = \begin{cases} e^{-r(T - t_{m-1})} R, & x \geq h, \\ c(x, t_{m-1}), & x < h. \end{cases}$$

- That is,

$$c(x, t_{m-1}) \approx \hat{c}(x, t_{m-1}) = e^{-r\Delta t} \sum_{k=0}^{N-1} F_k(x) V_k(t_m),$$

- $F_k(x) = \operatorname{Re} \left[\varphi \left(\frac{k\pi}{b-a}; x \right) e^{-\frac{ik\pi a}{b-a}} \right];$
- $V_k(t_m)$ are the Fourier-cosine coefficients of $v(y, t_m)$:

$$V_k(t_m) = \frac{2}{b-a} \int_a^b v(y, t_m) \cos \left(\frac{k\pi(y-a)}{b-a} \right) dy.$$

- $\varphi(\omega; x)$ is the conditional characteristic function of y given x ;
- N is the number of COS terms;
- $[a, b]$ defines the truncation range.

Pricing Barrier Options by the COS Method (2/8)

- Keep in mind that once the continuation value at time t_{m-1} is approximated, the option value at t_{m-1} becomes another approximation:

$$v(x, t_{m-1}) \approx \hat{v}(x, t_{m-1}) = \begin{cases} e^{-r(T-t_{m-1})} R, & x \geq h, \\ \hat{c}(x, t_{m-1}), & x < h. \end{cases}$$

- As a result, $V_k(t_{m-1})$ needed for $\hat{c}(x, t_{m-1})$ carries another layer of approximation:

$$V_k(t_{m-1}) \approx \hat{V}_k(t_{m-1}) = \frac{2}{b-a} \int_a^b \hat{v}(y, t_{m-1}) \cos\left(\frac{k\pi(y-a)}{b-a}\right) dy,$$

$$\hat{c}(x, t_{m-1}) \approx e^{-r\Delta t} \sum_{k=0}^{N-1} ' F_k(x) \hat{V}_k(t_m),$$

$$F_k(x) = \operatorname{Re} \left[\varphi\left(\frac{k\pi}{b-a}; x\right) e^{-\frac{ik\pi a}{b-a}} \right].$$

- The key is to solve the coefficients $\hat{V}_k(t_m)$ by backward induction.

Pricing Barrier Options by the COS Method (3/8)

- Calculation of $\hat{V}_k$:

$$\begin{aligned}\hat{V}_k(t_m; i = 1) &= \frac{2}{b-a} \int_a^b \hat{v}(y, t_m) \cos\left(\frac{k\pi(y-a)}{b-a}\right) dy \\ &= \underbrace{\frac{2e^{-r(T-t_{m-1})}R}{b-a} \int_h^b \cos\left(\frac{k\pi(y-a)}{b-a}\right) dy}_{\Psi_k(h, b)} + \underbrace{\frac{2}{b-a} \int_a^h \hat{c}(y, t_m; i = 1) \cos\left(\frac{k\pi(y-a)}{b-a}\right) dy}_{\hat{C}_k(a, h, t_m; i=1)}.\end{aligned}$$

- Inserting the continuation value $\hat{c}(x, t_m; i = 1)$ into the integral,

$$\begin{aligned}\hat{C}_k(x_1, x_2, t_m; i = 1) &= \frac{2}{b-a} \int_{x_1}^{x_2} \hat{c}(x, t_m; i = 1) \cos\left(\frac{k\pi(x-a)}{b-a}\right) dx \\ &= e^{-r\Delta t} \operatorname{Re} \left[\sum_{j=0}^{N-1} {}' \varphi_{\text{Levy}}\left(\frac{j\pi}{b-a}\right) V_j(t_{m+1}) M_{k,j}(x_1, x_2) \right],\end{aligned}$$

with

$$M_{k,j}(x_1, x_2) = \frac{2}{b-a} \int_{x_1}^{x_2} e^{ij\pi(x-a)/(b-a)} \cos\left(\frac{k\pi(x-a)}{b-a}\right) dx,$$

which can be solved analytically.

- $$\begin{aligned}\hat{c}(x, t_m; \mathbf{i}_m = \mathbf{1}) &= \mathbb{E}[v(t_{m+1}); i_m = 1] \\ &= p(i_{m+1} = 1) \cdot \mathbb{E}[v(t_{m+1}) \mid i(t_{m+1}) = 1, i(t_m) = 1] \\ &\quad + p(i_{m+1} = 2) \cdot \mathbb{E}[v(t_{m+1}) \mid i(t_{m+1}) = 2, i(t_m) = 1] .\end{aligned}$$

- $$\mathbb{E}[v(t_{m+1}) \mid i(t_{m+1}) = 2, i(t_m) = 1] = \int y f(y \mid i_{m+1}, i_m) dy = \int y f(y \mid i_{m+1}, i_m) dy .$$

Pricing Barrier Options by the COS Method (4/8)

- Summary so far:

$$\hat{V}_k(t_m) = \hat{C}_k(a, h, t_m) + e^{-r(T-t_{m-1})} \frac{2R \Psi_k(h, b)}{b-a}.$$

- Terminal condition is given by $\hat{V}_k(t_M) = V_k(t_M)$:

- For $h < 0$:

$$V_k(t_M) = \begin{cases} \frac{2R \Psi_k(h, b)}{b-a}, & \text{call,} \\ G_k(a, h) + \frac{2R \Psi_k(h, b)}{b-a}, & \text{put.} \end{cases}$$

- For $h \geq 0$:

$$V_k(t_M) = \begin{cases} G_k(0, h) + \frac{2R \Psi_k(h, b)}{b-a}, & \text{call,} \\ G_k(a, 0), & \text{put.} \end{cases}$$

Pricing Barrier Options by the COS Method (5/8)

- The calculation of $\hat{C}_k$ boils down to solving matrix-vector products.
- Define

$$\mathbf{u} := \{u_j\}_{j=0}^{N-1}, \quad u_j := \varphi\left(\frac{j\pi}{b-a}\right) V_j(t_{m+1}), \quad u_0 = \frac{1}{2} \varphi(0) V_0(t_{m+1}).$$

- The element $M_{k,j}(x_1, x_2)$ splits into a cosine matrix $M_{k,j}^c$ and a sine matrix $M_{k,j}^s$:

$$M_{k,j}^c = \begin{cases} \frac{(x_2 - x_1)\pi i}{b-a}, & k = j = 0, \\ \frac{\exp(i(j+k)\frac{(x_2-a)\pi}{b-a}) - \exp(i(j+k)\frac{(x_1-a)\pi}{b-a})}{j+k}, & \text{otherwise.} \end{cases}$$

$$M_{k,j}^s = \begin{cases} \frac{(x_2 - x_1)\pi i}{b-a}, & k = j, \\ \frac{\exp(i(j-k)\frac{(x_2-a)\pi}{b-a}) - \exp(i(j-k)\frac{(x_1-a)\pi}{b-a})}{j-k}, & k \neq j. \end{cases}$$

- Then

$$\hat{C}(x_1, x_2, t_m) = \frac{e^{-r\Delta t}}{\pi} \operatorname{Im}\{(\mathcal{M}_c + \mathcal{M}_s) \mathbf{u}\}.$$

Pricing Barrier Options by the COS Method (6/8)

- $\mathcal{M}_c$ is a **Hankel matrix**:

$$\mathcal{M}_c = \begin{bmatrix} m_0 & m_1 & m_2 & \cdots & m_{N-1} \\ m_1 & m_2 & \cdots & \cdots & m_N \\ \vdots & & & & \vdots \\ m_{N-2} & m_{N-1} & \cdots & & m_{2N-3} \\ m_{N-1} & \cdots & & m_{2N-3} & m_{2N-2} \end{bmatrix}_{N \times N}.$$

- $\mathcal{M}_s$ is a **Toeplitz matrix**:

$$\mathcal{M}_s = \begin{bmatrix} m_0 & m_1 & \cdots & m_{N-2} & m_{N-1} \\ m_{-1} & m_0 & m_1 & \cdots & m_{N-2} \\ \vdots & & \ddots & & \vdots \\ m_{2-N} & \cdots & m_{-1} & m_0 & m_1 \\ m_{1-N} & m_{2-N} & \cdots & m_{-1} & m_0 \end{bmatrix}_{N \times N}.$$

- Property of Toeplitz matrices.* The product $\mathcal{M}_s \mathbf{u}$ equals the first N elements of $\mathbf{m}_s \circledast \mathbf{u}_s$ with

$$\mathbf{m}_s = [m_0, m_{-1}, m_{-2}, \dots, m_{1-N}, 0, m_{N-1}, m_{N-2}, \dots, m_1]^T.$$

- Property of Hankel matrices.* The product $\mathcal{M}_c \mathbf{u}$ equals the first N elements of $\mathbf{m}_c \circledast \mathbf{u}_c$ (in reversed order), with

$$\mathbf{m}_c = [m_{2N-1}, m_{2N-2}, \dots, m_1, m_0]^T, \quad \mathbf{u}_c = [0, \dots, 0, u_0, u_1, \dots, u_{N-1}]^T.$$

Pricing Barrier Options by the COS Method (7/8)

- Recall the discrete Fourier transform:

$$\mathcal{D}(\mathbf{x})_k = \sum_{n=0}^{N-1} x_n e^{j \frac{2\pi}{N} kn}.$$

- For Toeplitz/Hankel-vector products, use circulant embedding:

$$\mathbf{x} \circledast \mathbf{y} = \mathcal{D}^{-1}\{\mathcal{D}(\mathbf{x}) \cdot \mathcal{D}(\mathbf{y})\}.$$

- Use the FFT to reduce the computational complexity from $O(N^2)$ to log-linear, i.e. $O(N \log_2 N)$.
- Overall complexity of the COS method: $O((M-1) N \log_2 N)$, with M monitoring dates and N COS terms.

Pricing Barrier Options by the COS Method (8/8)

- **Summary of the algorithm:**

- The terminal condition $V_k(t_M)$ is known: $\hat{V}_k(t_M) = V_k(t_M)$.
- For $m = M, M-1, \dots, 2$:
 - Use $\hat{V}_k(t_m)$ to obtain the continuation value $\hat{c}(x, t_{m-1})$ at t_{m-1} by COS:

$$\hat{c}(x, t_{m-1}) = e^{-r\Delta t} \sum_{k=0}^{N-1} {}' F_k(x) \hat{V}_k(t_{m-1}), \quad F_k(x) = \operatorname{Re} \left[\varphi \left(\frac{k\pi}{b-a}; x \right) e^{-\frac{ik\pi a}{b-a}} \right].$$

- Update $\hat{C}_k(t_{m-1})$ via FFT.
 - Update $\hat{V}_k(t_{m-1})$.
- Finally, calculate the option price at the initial time:

$$\hat{v}(x, t_0) = e^{-r\Delta t} \sum_{k=0}^{N-1} {}' F_k(x) \hat{V}_k(t_1).$$

The Truncation Range

- The determination of the truncation range $[a, b]$ is important.
 - An interval that is too small leads to a significant integration-range truncation error.
 - A very large interval requires a large N to achieve a given accuracy.
- As a rule of thumb, define the integration range by

$$[a, b] = \left[(c_1 + x_0) - L\sqrt{c_2 + \sqrt{c_4}}, (c_1 + x_0) + L\sqrt{c_2 + \sqrt{c_4}} \right],$$

where $x_0 = \ln(S_0/K)$, L depends on a user-defined tolerance level TOL , and $c_1, \dots, c_4$ are the cumulants based on the characteristic function of the underlying process.

Testing Results

Table 5 Errors and CPU times for pricing monthly-monitored barrier options under the CGMY model (Test no. 4)

Option type	Ref. val.	N	Time (ms)	Error	Ratio
DOP	2.339381026	2^4	2.8	$2.23\text{e-}1$	–
		2^5	2.7	$1.98\text{e-}2$	2.6
		2^6	3.4	$3.23\text{e-}4$	2.0
		2^7	4.6	$7.20\text{e-}9$	2.3
DOC	9.155070561	2^4	2.7	$5.06\text{e-}2$	–
		2^5	2.9	$5.67\text{e-}3$	1.7
		2^6	3.3	$1.99\text{e-}4$	1.6
		2^7	4.7	$5.55\text{e-}9$	2.2
UOP	6.195603554	2^4	3.0	$5.58\text{e-}2$	–
		2^5	2.9	$8.98\text{e-}3$	1.6
		2^6	3.6	$1.96\text{e-}4$	1.8
		2^7	4.8	$2.23\text{e-}8$	2.1
UOC	1.814827593	2^4	2.8	$3.38\text{e-}1$	–
		2^5	2.8	$1.24\text{e-}2$	4.0
		2^6	3.5	$3.45\text{e-}6$	2.9
		2^7	4.7	$1.93\text{e-}8$	1.4

For Heston's Model

- The same approach can be applied to Heston's model.
- The expectation integral becomes a two-dimensional integral:
 - 1-D COS for the log-stock dimension + numerical integration on the (log-)variance dimension; or
 - 2-D COS for recovering the joint density of log-stock and variance.

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